WRFall Quick Reference

`wrfall_install.pl` v1.4.2 · one-page commands for installing WRF/WPS, creating a domain, downloading data and running a forecast.

1. Daily workflow (4 commands)

```
./wrfall --mkdomain ethiowrf                          # once per domain
./wrfall --getdata gfs --date 20261016 --end 18       # download data
RUN_GEOGRID=1 ./wrfall --run 20261016 00 18           # first run (builds geo_em)
./wrfall --run 20261016 00 18                         # later runs
```

Run these from inside the domain folder (for example `~/Build_WRF/ethiowrf`). From anywhere else, add `--domain ~/Build_WRF/ethiowrf`. Results are in `wrf_output/<YYYYMMDDCC>/`.

2. Install and maintain

Task	Command
Show the main help	`wrfall_install.pl --help`
List all flags	`wrfall_install.pl --flags`
List releases you can install	`wrfall_install.pl --install list`
Install ARW (default build)	`wrfall_install.pl --install -arw`
Install a chosen release with all builds	`wrfall_install.pl --install 4.7.0 -all`
Preview without doing anything	add `--dryrun`
Answer yes to all questions	add `--allyes`
Resume an interrupted build	add `--continue`
Update to a newer release	`wrfall_install.pl --update`
Install extras	`--addpack domwiz,scripts,cdsapi`
Add terrestrial data	`--addgeog chem` (or `mandatory`, `fire`)
Skip the base `WPS_GEOG` download	`--nogeog`

3. Build types

Flag	Builds
`-arw`	WRF-ARW (standard)
`-chem`	WRF-Chem

Flag	Builds
`-hydro`	WRF-Hydro
`-fire`	WRF-Fire (real-data)
`-fireideal`	WRF-Fire ideal `em_fire` cases
`-all`	All of the above

4. Download data (`--getdata`)

Option	Meaning	Default
`gfs` | `ecmwf` | `era5`	Data source	required
`--date YYYYMMDD`	Start date (UTC)	today
`--cycle 00|06|12|18`	Model cycle	`00`
`--start H` / `--end H`	First / last forecast hour	`0` / `120` (ERA5: `21`)
`--step H`	Hours between files	`3`
`--margin DEG`	Extra degrees around the domain	`1.0`
`--bbox W E S N`	Set the area by hand	read from `namelist.wps`
`--domain DIR`	Domain folder	current folder
`--force`	Download again	off
`--dryrun`	Preview only	off

Files are saved in `GFS_Input/`, `ECMWF_Input/` or `ERA5_Input/` under a `YYYYMMDDCC` sub-folder. Existing files are skipped, so you can safely run the command again.

5. Run WRF (`--run`)

```
./wrfall --run <YYYYMMDD> <CC> <HH> [gfs|ecmwf|era5]
./wrfall --run auto                 # today, 00 UTC, 72 hours
```

Argument	Allowed values
Date	`YYYYMMDD`
Cycle	`00`, `06`, `12`, `18`
Forecast hours	1 to 72

Pipeline: set times → (geogrid) → link GRIB and Vtable → `ungrib` → `metgrid` → update `namelist.input` → `real.exe` → `wrf.exe`.

Setting	Effect	Default

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`RUN_GEOGRID=1`	Also run `geogrid.exe`	skipped
`WRF_NP=N`	MPI ranks for `wrf.exe`	8
`REAL_NP=N`	MPI ranks for `real.exe`	2

Example: `WRF_NP=16 RUN_GEOGRID=1 ./wrfall --run 20261016 00 18`

6. Domain folder map

Path	Contents
`namelist.wps`, `namelist.input`	Your settings
`GFS_Input/YYYYMMDDCC/`	Downloaded GRIB files
`wrf_output/YYYYMMDDCC/`	Final `wrfout_*` files
`wrf_output/geogrid`, `ungrib`, `metgrid`	Archived step products
`rsl.error.0000`, `rsl.out.0000`	Run logs (check `rsl.error.0000` first on a failure)

7. Useful flags

Flag	Purpose
`--homedir DIR`	Install location (default `$WRF_HOME` or `~/Build_WRF`)
`--repodir DIR`	Use a local tarball repository instead of the network
`--dl-jobs N`	Parallel download connections per file (default 8)
`--curl` / `--wget`	Force a download tool
`--proxy HOST:PORT`	Use a proxy (proxies are cleared by default)
`--force`	Rebuild WRF/WPS even if present
`--force-libs`	Also rebuild the support libraries
`--scour`	Delete old release folders after a successful install or update
`--import DIR`	Import domains from an older installation
`--wrf-opt N --wps-opt N`	Override the configure choices (defaults 34 and 3)
`--debug`, `--version`	Troubleshooting and version info

8. Quick fixes

Problem	Fix
GFS file shows `[FAIL]`	Data not online yet or too old; wait or use another cycle
`GRIB input directory not found`	Run `--getdata` with the same date and cycle
No `geo_em` file	Run once with `RUN_GEOGRID=1`
`real.exe failed`	Read `rsl.error.0000`; check the dates and forecast hours
`wrf.exe` stops early	Read `rsl.error.0000`; try a smaller `time_step`
Downloads blocked by a firewall	Add `--proxy HOST:PORT`

Remember: the date, cycle and hours in `--run` must match the data you downloaded with `--getdata`.